

Evaluating a UI Design

Expert inspection methods

Cognitive Walkthrough

Heuristic Evaluation

Benefits:

Applicable early in the design life cycle –can be
used with paper design, prototype or full system

Based on simple methodologies – not expensive

Notes: Evaluators should not be a member of design team

For best results, 3-8 evaluators required

Cognitive Walkthrough

- Begin with
 - a detailed *interaction design scenario* including complete sequence of actions, and screen design, menu's etc., and
 - a user profile
- Evaluators “step through” the sequence of actions
- At each step, analyze whether the user would be likely to perform the correct action, by asking 4 questions

Cognitive Walkthrough (cont.)

Questions:

1. Will user's current goal match the effect of this action? (i.e., will the user know *what* to do next?)
1. Is the affordance for this action visible?
2. Will users be able to identify this action with its desired effect? (can the user see *how* to do it?)
3. After the action, will users understand the feedback they receive?

Note how this is related to Norman's "Gulf of Execution" and "Gulf of Evaluation"?

User Interface Design Principles (“Heuristics”)

Nielsen: Original 10 Usability Heuristics (1990)

Simple and natural dialogue

Clearly marked exits

Speak the user’s language

Shortcuts

Minimize memory load

Good error messages

Consistency

Prevent errors

Feedback

Help and documentation

Previously, published guidelines had hundreds or thousands of rules

Nielsen: Revised 10 Usability Heuristics (based on extensive empirical testing)

*Visibility of system status
(i.e. feedback)

Match between system and the
real world (speak the user's
language)

*User control and freedom
(undo, redo, clear exits)

*Consistency

*Error prevention

Recognition not recall
(minimize memory load)

*Flexibility and efficiency
(includes shortcuts, macros)

Aesthetic and minimalist
design

*Help users diagnose and
recover from errors

Help and documentation

* Also included in Shneiderman's 8 golden rules

User Interface Principles/Heuristics (cont.)

Shneiderman's 8 Golden Rules

- *Strive for consistency
- *Enable shortcuts for frequent users
- *Offer informative feedback
- Design dialogs to yield closure
- *Offer error prevention and simple error handling
- *Permit easy reversal of actions
- *Support *user* locus of control
- *Reduce memory load

User Interface Principles/Heuristics (cont.)

Guidelines for reducing errors by ensuring complete and correct actions:

- Prefer menus and selection lists to commands and data entry

- Check entry fields for legal values

- Insert or signal begin-end of paired markers (such as parentheses)

- Automate important sequences of user actions (macros)

- Ensure correct commands (completion, etc.) ??

Q: What other heuristics and guidelines will help to prevent errors?

Heuristic Evaluation

Definition and goals:

Systematic inspection of a UI design for usability

Using a set of heuristics or principles

Purpose is to identify usability problems

General approach:

Conducted by a small number of evaluators (how many?)

Each evaluator inspects the interface independently

An observer may be present, to take notes and answer questions about the interface or the application/task

Result is a written report or observer's notes

(observer's notes may be ready sooner)

Output is *a list of usability problems* explained in terms of the guidelines that were violated

Heuristic Evaluation (cont.)

Procedures:

Typically 1-2 hours

Evaluator inspects UI elements and compares them with a list of principles/heuristics

Go through interface at least twice:

first, get acquainted with the system

then, evaluate specific UI elements:

- a. information organization and general task flow
- b. mechanisms provided (or lack of them)
- c. visual characteristics

Note the importance of evaluator expertise of 2 kinds:
usability expertise and application expertise

Heuristic Evaluation (cont.)

Severity ratings:

- 0 - this is not a usability problem
- 1 - cosmetic problem -- need not be fixed unless time permits
- 2 - minor usability problem
- 3 - major usability problem, should be given priority
- 4 - usability catastrophe -- imperative to fix before product can be released

Depends on:

Proportion of users who can be expected to experience the problem

Impact of the problem on the users who experience it

Heuristic Evaluation (cont.)

Contrast with user testing:

- in user testing, users not experts in UI design

- in heuristic evaluation, evaluators may not know much about the application domain and the tasks being performed

- user testing involves realistic tasks

- in user testing, hints and assistance are avoided

- in user testing, the observer does the analysis

- heuristic evaluation can take place at an early stage using:

 - paper mock-up designs

 - unstable prototypes

 - hints and advice replacing documentation and help

Heuristic Evaluation (cont.)

Generating fixes:

Some fixes can be identified easily, such as the need for new mechanisms, consistent appearance, or more feedback

Some problems require major re-design (confusing menu structure)

Group Debriefing - Brainstorming with several evaluators + designers